Legacy Architecture — Current State

Document Purpose

This document captures **how the system works today** — the complete call tracking and case management architecture built on CallTrackingMetrics (CTM), Twilio, Zoho, Flow Legal Management, Snowflake, and supporting systems. It serves as the baseline that any replacement must match or exceed before the legacy platform can be decommissioned.

Scope: Current state only. For identified inefficiencies and redesign opportunities, see ARCHITECTURE_OPPORTUNITIES.md. For the target-state RustPBX platform design, see ARCHITECTURE_VISION.md.

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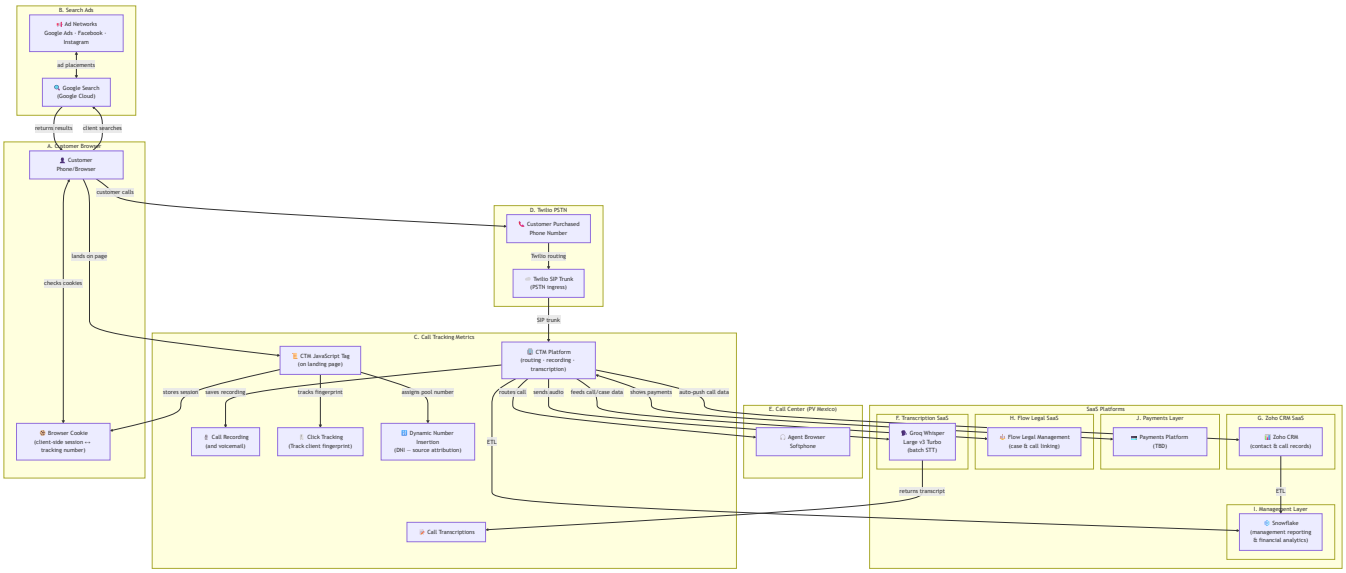
1. System Overview

The legacy architecture spans ten layers (A–J), from the customer's initial ad click through to case management in Flow Legal. CTM sits at the center as the integration hub — it receives calls from Twilio, routes them to agents, records and transcribes them, and pushes data downstream to Zoho CRM, Flow Legal Management, and Snowflake (the data warehouse for management reporting and financial analytics).



2. Customer Acquisition (Layers A–B)

2.1 Customer Phone/Browser (Layer A)

The customer journey begins on a mobile phone or desktop browser. The prospective customer — typically someone seeking legal services — initiates a search that eventually leads to a phone call with our call center.

Customer profiles:

- Mobile (majority): searching on iPhone/Android, may click-to-call directly from ad
- Desktop: searching on laptop/desktop, calls from cell phone after viewing landing page
- Returning visitor: may have previously visited the site, cookie may or may not persist

2.2 Google Search (Layer B)

Customers discover our services through search engines, predominantly Google Search. The typical query pattern is service + location (e.g., “personal injury lawyer near me”, “car accident attorney Houston”). These searches trigger paid ads from the ad networks described below, as well as organic search results pointing to our landing pages.

2.3 Ad Networks (Layer B)

Paid advertising across three platforms drives the majority of inbound call volume:

Platform	Ad Format	Tracking Parameter	Click ID
Google Ads	Search ads, display ads, call-only ads	`utm_source=google&utm_medium=cpc`	`gclid` (auto-appended)
Facebook Ads	News feed ads, story ads, lead forms	`utm_source=facebook&utm_medium=paid`	`fbclid` (auto-appended)
Instagram Ads	Feed ads, story ads, reels ads	`utm_source=instagram&utm_medium=paid`	`fbclid` (shared with Facebook)

Each platform appends its own click identifier to the landing page URL. CTM’s JavaScript captures these parameters to attribute the eventual phone call back to the originating ad click, campaign, and

keyword.

Google Ads call-only ads are a special case: the customer taps to call directly from the search results page without ever visiting a landing page. In this flow, Google forwards the call to a Google forwarding number, which routes to CTM. Attribution relies on Google's own call conversion tracking rather than CTM's DNI JavaScript.

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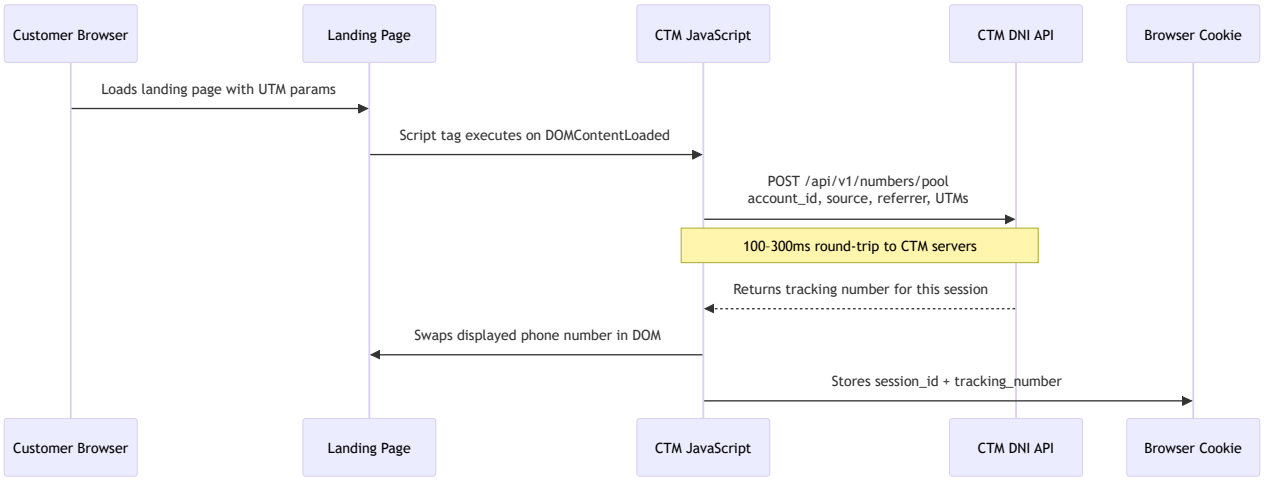
3. Click Tracking & Source Attribution (Layer C)

3.1 CTM JavaScript Tag (Layer C)

Every landing page includes a CTM-provided JavaScript snippet that loads on page render. This script performs two functions: tracking the visit and swapping the phone number.

How it works today:



CTM JavaScript snippet (typical):

```
<script type="text/javascript">
  var __ctm_loaded = false;
  (function() {
    var s = document.createElement('script');
    s.type = 'text/javascript';
    s.async = true;
    s.src = ('https:' == document.location.protocol ? 'https://' : 'http://')
            + 'ACCOUNT_ID.ctm.com/t.js';
    var p = document.getElementsByTagName('script')[0];
    p.parentNode.insertBefore(s, p);
  })();
</script>
```

The script identifies phone number elements on the page (by CSS class, data attribute, or content matching) and replaces the static business number with a dynamically assigned tracking number unique to that visitor session.

3.2 Dynamic Number Insertion — DNI (Layer C)

DNI is the mechanism that ties an ad click to a phone call. CTM maintains a pool of phone numbers and assigns one to each visitor session. When the customer eventually calls that number, CTM knows which ad, campaign, keyword, and landing page drove the call.

Number pool management:

Aspect	Current Behavior
Pool size	CTM manages the pool; numbers are provisioned through Twilio
Assignment	One tracking number per visitor session
Recycling	Number returns to pool after session expires (configurable timeout)
Shortage handling	If pool is exhausted, falls back to default business number (no attribution)
Number format	US local numbers matching the area code of the landing page’s target market

Attribution chain:

Cookie dependency: The entire attribution chain relies on the browser cookie persisting between the page visit and the phone call. This is fragile — cookies can be cleared, blocked by Safari ITP, stripped by ad blockers, or absent entirely in incognito mode. When the cookie is lost, the call still arrives at CTM but without source attribution, appearing as an “unattributed” or “direct” call.

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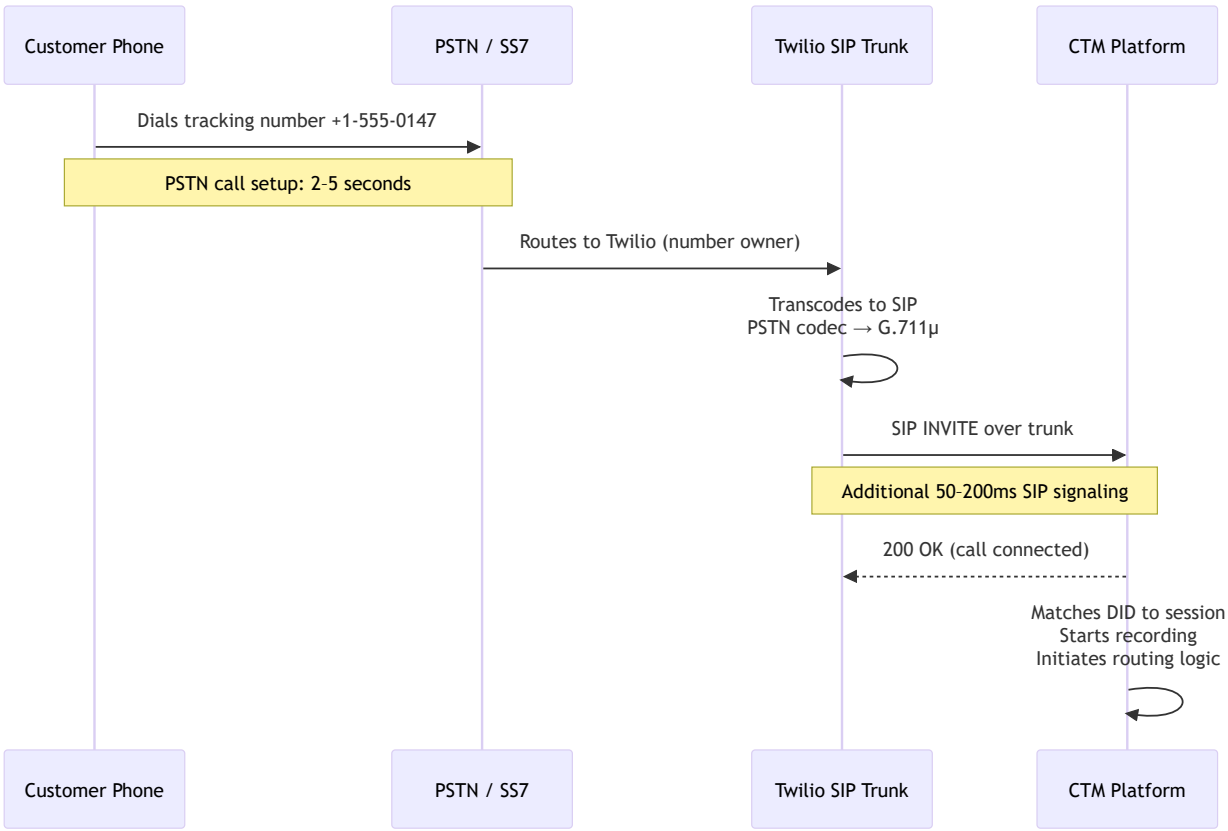
4. PSTN Call Path (Layer D)

When the customer dials the tracking number, the call traverses the public telephone network (PSTN) through Twilio’s infrastructure before reaching CTM.

4.1 Call Flow: Customer to CTM



4.2 Twilio SIP Trunking

Twilio serves as the PSTN-to-SIP bridge. CTM uses Twilio SIP trunking to receive inbound calls and make outbound calls.

Parameter	Value
Service	Twilio Elastic SIP Trunking
Inbound	Twilio owns the tracking numbers, routes to CTM via SIP trunk
Outbound	CTM sends SIP to Twilio for PSTN termination
Codec	G.711μ (PCMU) — Twilio transcodes from PSTN as needed
Recording	CTM handles recording (not Twilio)
Billing model	Per-minute: ~$0.0085/min inbound PSTN + ~$0.005/min SIP trunk
Combined cost	~$0.0135/min (double billing — PSTN leg + SIP leg)
Number porting	2–4 weeks to port numbers away from Twilio

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5. CTM Platform (Layer C)

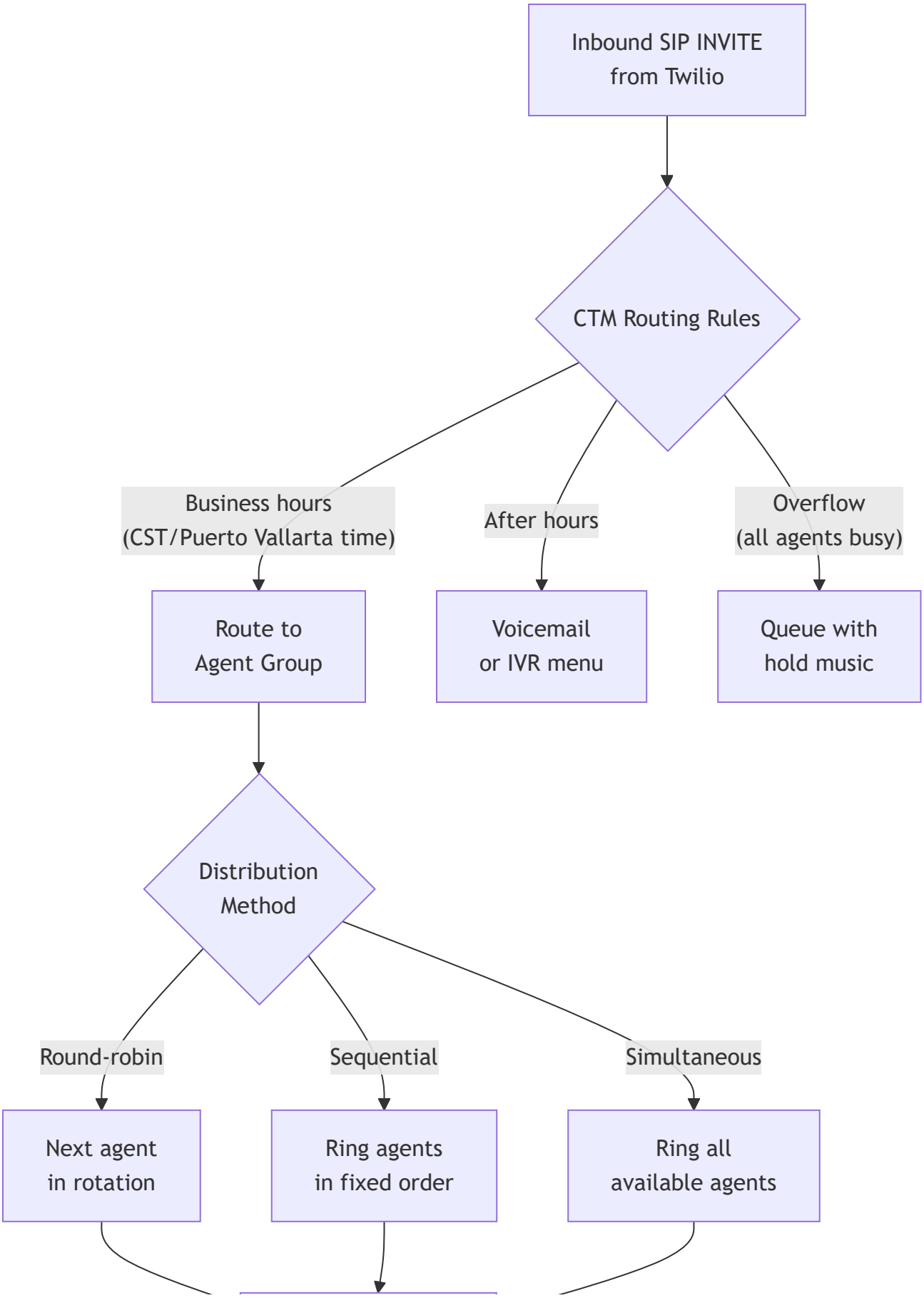
CallTrackingMetrics (CTM) is the central hub of the legacy architecture. It receives calls from Twilio, routes them to agents, records conversations, provides basic transcription, and pushes data to downstream systems.

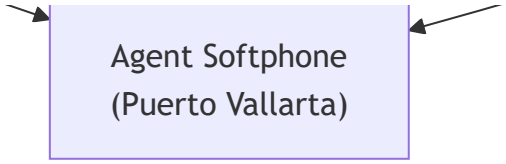
5.1 Core CTM Functions

Function	How It Works Today
Call routing	GUI-configured routing rules: time-of-day, geo, round-robin, sequential ring
IVR	Basic auto-attendant with DTMF menu trees
Call recording	Automatic recording of all calls; stored in CTM cloud
Transcription	CTM's built-in transcription (provider unknown, likely third-party ASR)
Call scoring	Manual agent scoring via CTM interface
Reporting	Web dashboard with call volume, duration, source attribution, agent performance
Number management	CTM manages tracking number pools provisioned via Twilio
Whisper/announce	Pre-connect message to agent ("This call is from Google Ads - PI campaign")

5.2 CTM Call Routing

When a call arrives from Twilio, CTM's routing engine determines which agent or group should receive it.


Inbound SIP INVITE from Twilio
CTM Routing Rules
Business hours (CST/Puerto Vallarta time)
After hours
Overflow (all agents busy)
Route to Agent Group
Voicemail or IVR menu
Queue with hold music
Distribution Method
Round-robin
Sequential
Simultaneous
Next agent in rotation
Ring agents in fixed order
Ring all available agents



5.3 Call Recording & Storage

CTM records all calls automatically. Recordings are stored in CTM's cloud infrastructure with no direct access to raw files.

Aspect	Current State
Format	Proprietary; accessible via CTM dashboard and API
Storage location	CTM cloud (vendor-controlled)
Retention	Per CTM account settings
Download	Manual download through CTM UI; API access with rate limits
Bulk export	Limited; subject to API rate limiting
Ownership	Recordings are in CTM's infrastructure — portability is a concern

5.4 CTM Billing

CTM adds its own per-minute charges on top of Twilio's charges, creating a two-layer billing structure:

Cost Layer	Rate	Monthly (50K min)
Twilio PSTN + SIP	~$0.0135/min	~$675
CTM platform fee	Per-minute + subscription	Varies by plan
Total carrier + platform	~$0.02+/min	**~$1,000+**

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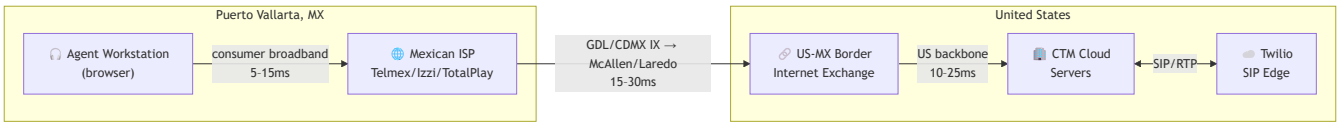
6. Call Center Agents (Layer E)

6.1 Agent Location: Puerto Vallarta, Mexico

The call center operates from **Puerto Vallarta, Jalisco, Mexico**. All agents connect to CTM from this location, which introduces cross-border networking considerations.

Network topology:



Latency profile (Puerto Vallarta → CTM):

Segment	Typical RTT
Agent → Mexican ISP PoP	5–15ms
Mexican ISP → US border crossing	15–30ms
US border → CTM cloud	10–25ms
Total round-trip	**30–70ms**

This is within acceptable limits for voice (ITU-T G.114 recommends < 150ms one-way), but leaves limited headroom for packet loss and jitter on consumer-grade ISPs.

6.2 Agent Experience

Agents interact with CTM through a browser-based interface. The current agent experience has several limitations that affect productivity.

What agents have today:

Capability	Status	Notes
Browser softphone	✅	CTM's built-in WebRTC softphone
Call controls (answer, hold, transfer, hangup)	✅	Standard CTM interface
Caller ID / source display	✅	CTM shows tracking source before connect
Whisper message	✅	Pre-connect audio: "Google Ads - PI National"
Call recording indicator	✅	Visible in CTM dashboard
Queue visibility	⚠️ Limited	Basic queue stats, no real-time agent dashboard
Live transcription	❌	Not available during calls
AI coaching / suggestions	❌	Not available
Real-time sentiment	❌	Not available
Dynamic call scripts	❌	Static scripts only (no context adaptation)
Supervisor live monitoring	⚠️ Limited	Listen-only; no whisper or barge
Post-call disposition	✅	Manual entry after each call
CRM screen pop	⚠️ Limited	Basic integration; no form pre-fill

6.3 Internet Reliability Concerns

Agents rely on consumer-grade Mexican ISPs with no dedicated circuits. This creates reliability risks:

ISP	Service	Known Issues
Telmex/Infinitum	DSL/Fiber	Frequent micro-outages, congestion during peak hours
Izzi	Cable	Speed fluctuations, regional outages
TotalPlay	Fiber	Better reliability but limited availability
Failover	Telcel 4G/5G mobile hotspot	Manual switch; adequate for voice but variable latency

Current failover strategy: Largely manual. If an agent's internet drops during a call, the call is lost. There is no automatic device failover, no ICE restart capability, and no secondary ISP auto-switching at the network level.

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7. Call Transcription (Layer F)

Call transcription operates in two tiers: CTM's built-in transcription and a supplemental Groq Whisper pipeline for higher-accuracy results.

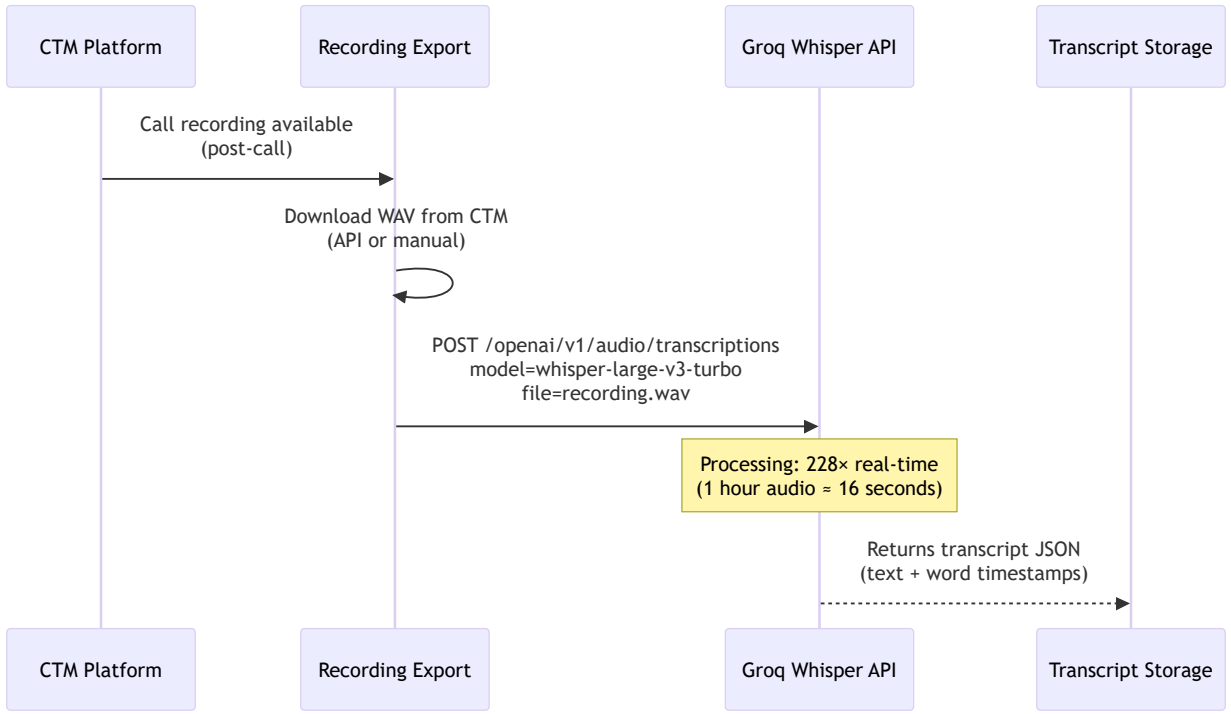
7.1 CTM Native Transcription

CTM provides post-call transcription as a platform feature. Transcripts appear in the CTM dashboard after call completion.

Aspect	Current State
Timing	Post-call only (no live transcription)
Latency	1–15 minutes after call ends
Accuracy	Moderate — CTM's ASR provider is not disclosed
Speaker diarization	Basic (caller vs. agent)
Searchability	Within CTM dashboard only
Export	Via CTM API (rate-limited)
Custom vocabulary	Not available
Languages	English primarily

7.2 Groq Whisper Large v3 Turbo (Supplemental)

For higher-accuracy transcription, call recordings are processed through **Groq Whisper Large v3 Turbo** — a hosted inference endpoint running OpenAI's Whisper Large v3 Turbo model on Groq's LPU hardware.



Groq Whisper specs:

Parameter	Value
Model	`whisper-large-v3-turbo`
API endpoint	`https://api.groq.com/openai/v1/audio/transcriptions`
API compatibility	OpenAI-compatible (drop-in replacement)
Processing speed	228× real-time
Cost	$0.04/hour ($0.00067/min)
Monthly cost (50K min/mo)	~$33
Max file size	25 MB per request
Word-level timestamps	✅
Speaker diarization	❌ Not built-in (Whisper limitation)
Languages	100+ (Whisper multilingual)
Custom vocabulary	Prompt-only (224 token limit)
Accuracy (WER)	~10–12% (Whisper v3 Turbo baseline)

Current workflow limitations:

- Recordings must be exported from CTM before Groq processing (manual or scripted)
- No real-time / streaming transcription (batch only)
- No automatic pipeline — requires orchestration outside CTM
- Two transcripts exist (CTM's and Groq's) with no single source of truth
- No speaker diarization without additional post-processing

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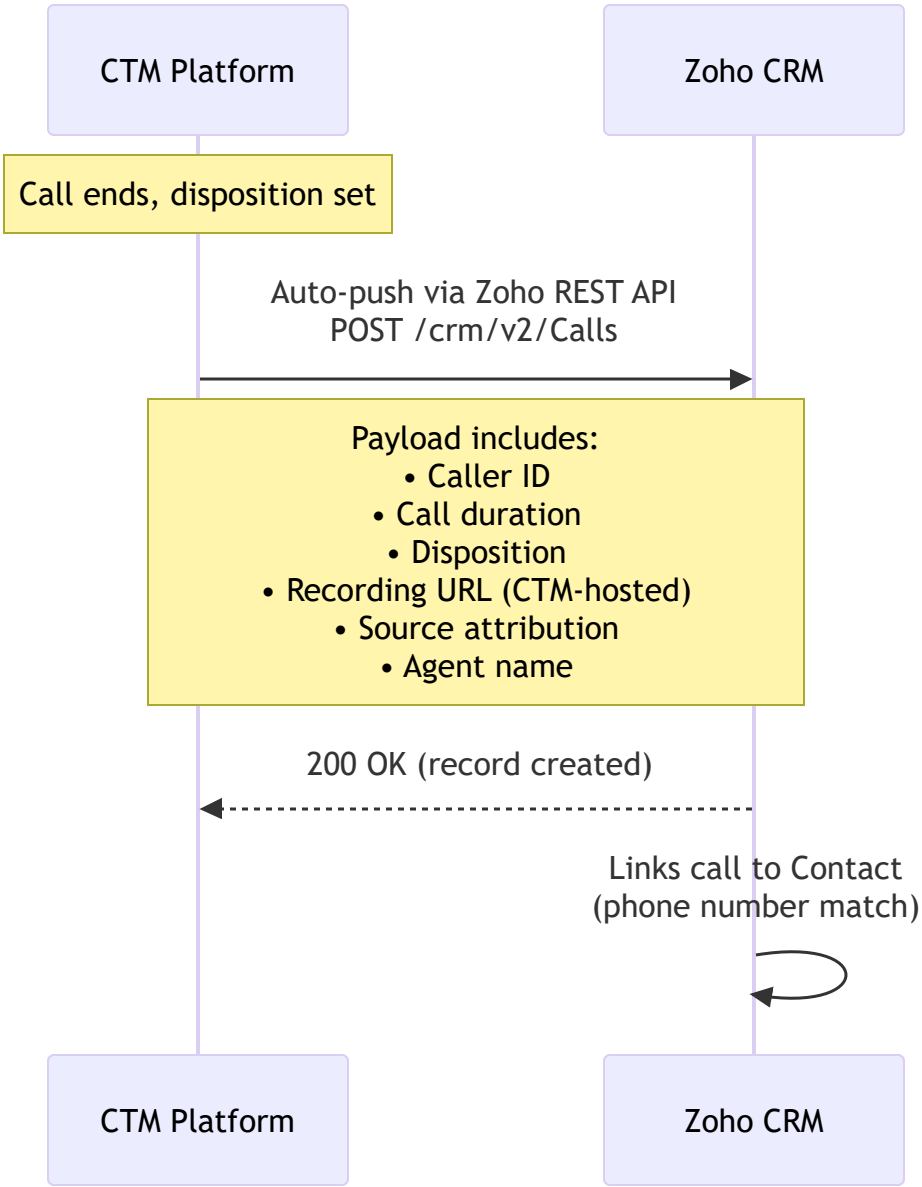
8. Zoho CRM Integration (Layer G)

CTM auto-pushes call data to Zoho CRM after each call completes. This is the primary CRM where contact records, call history, and lead information are maintained.

8.1 Integration Flow



8.2 Data Pushed to Zoho

Field	Source	Notes
Caller phone number	CTM call record	Used to match/create Contact
Call duration	CTM call record	
Call type	CTM	Inbound / Outbound
Disposition	Agent (manual)	Set by agent post-call
Recording URL	CTM-hosted link	Plays back through CTM; breaks if CTM is decommissioned
Source / campaign	CTM attribution	From DNI session tracking
Agent name	CTM	Who handled the call
Call timestamp	CTM	When the call occurred
CTM transcript	CTM (if enabled)	CTM's native transcription text

8.3 Zoho CRM Role

Zoho serves as the contact and lead management system:

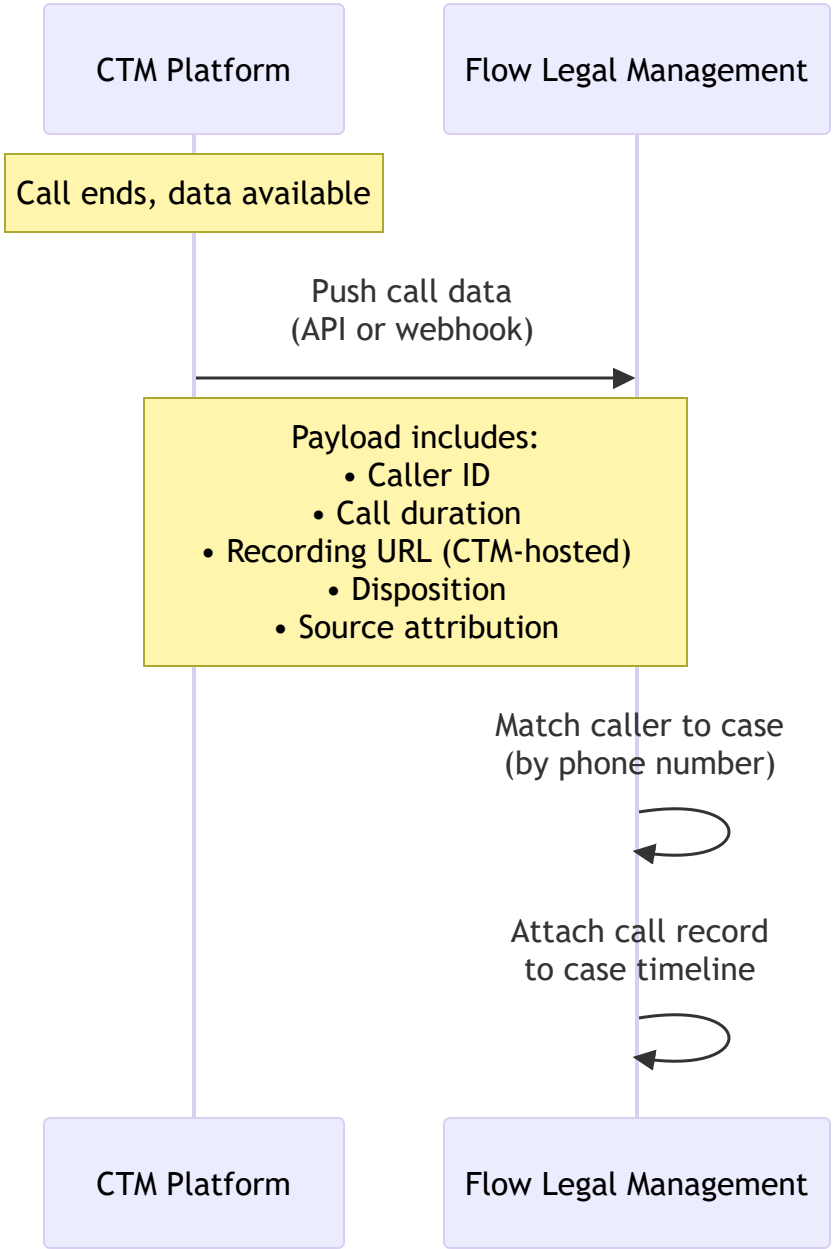
Function	How It's Used
Contact records	Phone number, name, email, lead source
Call history	Log of all calls associated with a contact
Lead tracking	New callers create leads; repeat callers link to existing contacts
Pipeline management	Move leads through stages (New → Qualified → Retained → Closed)
Reporting	Call volume by source, conversion rates, agent performance
Team access	Agents and managers access Zoho for contact context

> **Critical dependency:** Recording URLs in Zoho point to CTM-hosted files. If CTM is decommissioned, all historical recording links in Zoho will break unless recordings are migrated and URLs are updated.

9. Flow Legal Management Integration (Layer H)

CTM feeds call and case data to Flow Legal Management, the system used to manage legal cases and client matters. This integration connects incoming calls to active cases, enabling case managers to track client communication history.

9.1 Integration Flow



9.2 Data Pushed to Flow Legal

Field	Source	Notes
Caller phone number	CTM	Used to match to existing case/client
Call duration	CTM	
Recording URL	CTM-hosted	Same CTM dependency as Zoho
Disposition	Agent (manual)	
Call timestamp	CTM	
Source attribution	CTM	Ad source that drove the call

9.3 Flow Legal's Role

Function	How It's Used
Case management	Track legal matters from intake through resolution
Client communication log	All calls linked to the relevant case
Document management	Case files, correspondence, recordings
Task management	Follow-ups, deadlines, assignments
Intake workflow	New callers can trigger intake case creation

> **Critical dependency:** Like Zoho, recording links in Flow Legal point to CTM. CTM decommissioning requires recording migration and URL rewriting for both downstream systems.

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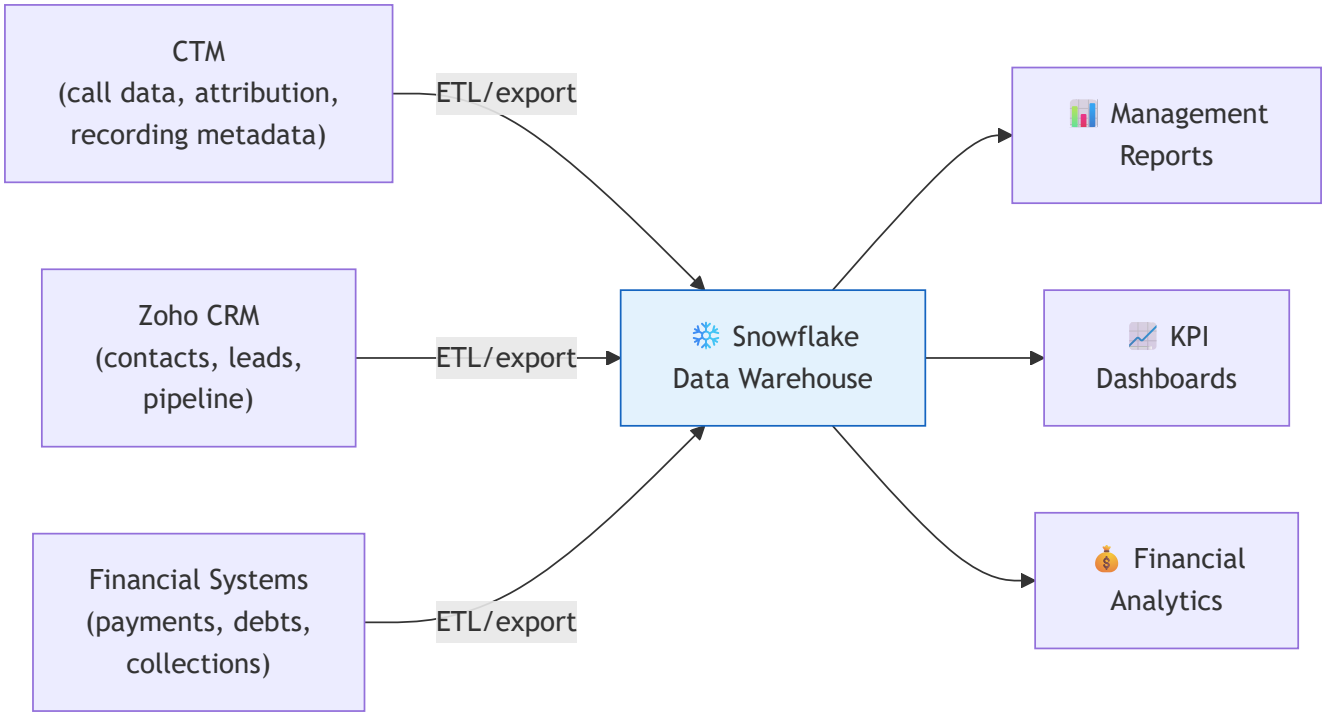
10. Snowflake — Management Reporting & Analytics (Layer I)

10.1 Role in the Legacy Architecture

Snowflake serves as the data warehouse for management reporting, aggregating data from multiple upstream systems into a single analytics layer.



10.2 Data Sources

Source	Data Fed to Snowflake
CTM	Call volumes, duration, disposition, source attribution, agent performance
Zoho CRM	Lead pipeline, contact history, conversion rates
Financial/payments	Client payments, outstanding debts, collections status

10.3 Reporting Capabilities

Snowflake enables cross-system queries that no individual vendor can provide on its own — for example, correlating ad spend attribution (CTM) with case conversion rates (Zoho) and revenue collected (financial data).

Report Category	Example Queries
Call center performance	Calls per agent, average handle time, disposition rates
Marketing attribution	Cost per lead by campaign, conversion rate by ad source
Financial	Revenue per case type, outstanding debt aging, collection rates
Cross-system	Ad spend → call → case → payment (full funnel ROI)

> **Vendor dependency:** Snowflake queries depend on data exports from CTM and Zoho. Replacing CTM requires redirecting the call data ETL pipeline. Replacing Zoho requires redirecting the CRM data pipeline. Both pipelines must be replicated or replaced before Snowflake (or its successor) can produce accurate reports.

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11. Payments (Layer J)

The payments layer is planned but not yet implemented. The specific platform (e.g., LawPay, Stripe, or a custom integration) is TBD. This layer will handle client payments, billing, and collections data — feeding into the management reporting layer (Snowflake / Layer I) for full-funnel ROI analysis from ad spend through to revenue collected.

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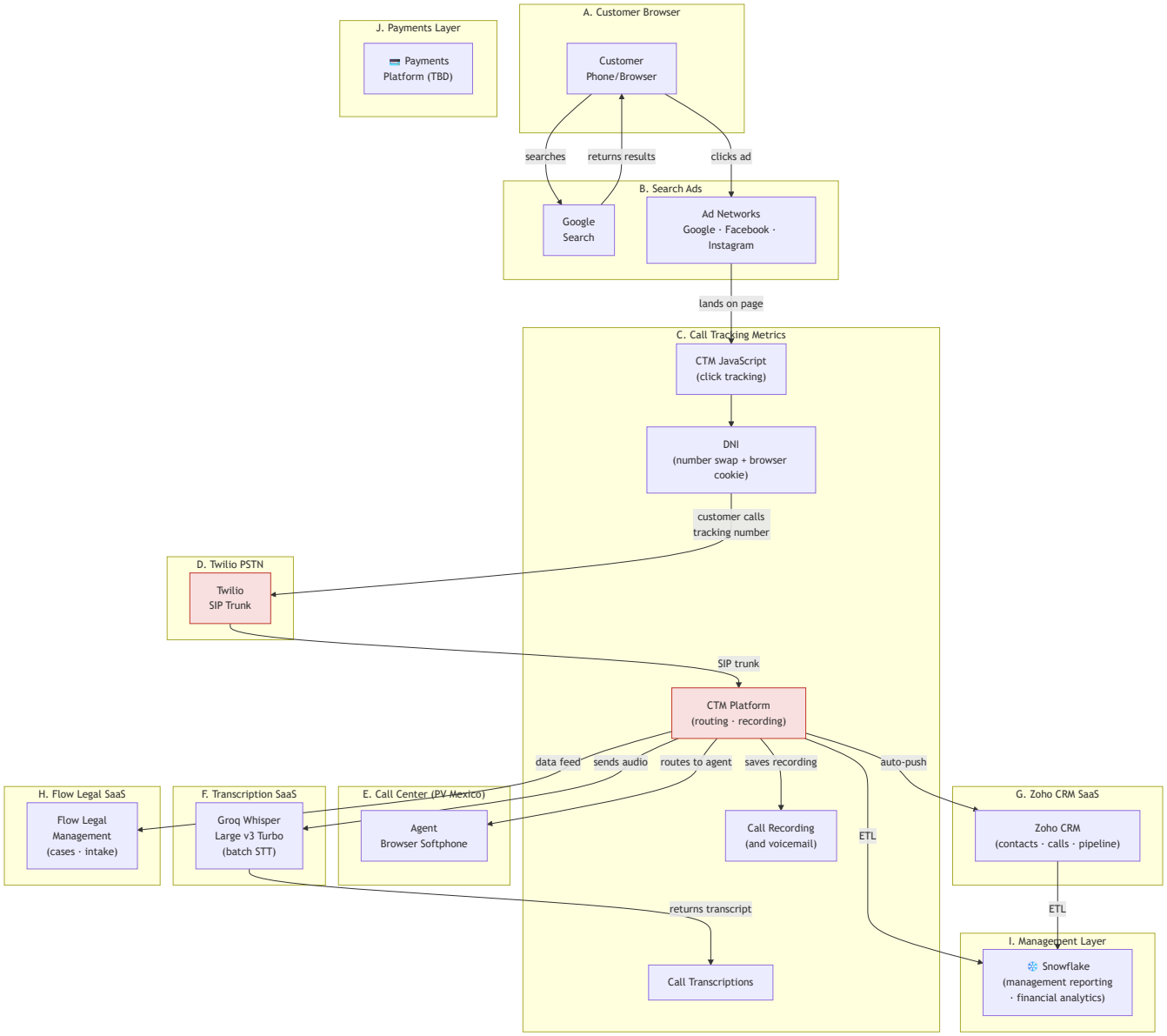
12. End-to-End Data Flow Summary

The complete legacy data flow, showing how information moves through all ten layers (A–J):



> **The red nodes (CTM + Twilio) are the components targeted for replacement.** CTM is the single point of failure for routing, recording, transcription, and all downstream integrations. Twilio is the sole PSTN carrier with 2–4 week number porting timelines.

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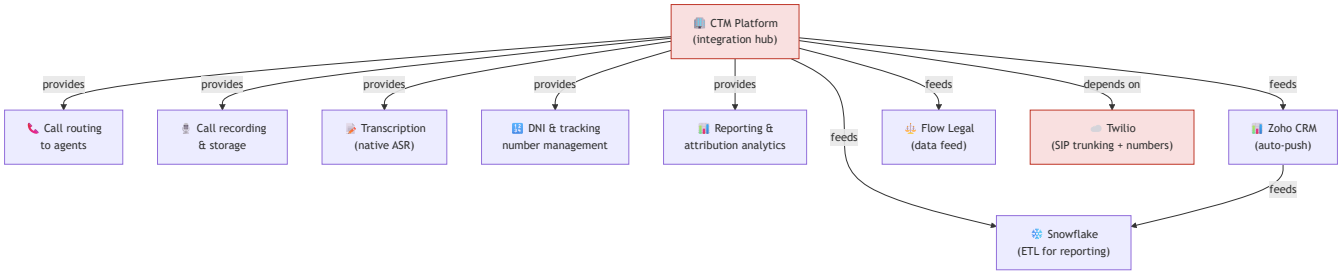
13. Technology Stack (Legacy)

Layer	Technology	Role
A. Customer Browser	Mobile/desktop browser	Customer search, ad click, page visit
B. Search Ads	Google Ads, Facebook Ads, Instagram Ads	Customer acquisition, paid search/social
C. Call Tracking	CTM JavaScript tag, CTM Platform	DNI, session tracking, source attribution, routing, recording
C. Session Storage	Browser cookies (via CTM JS)	Links page visit to phone call
D. PSTN Carrier	Twilio Elastic SIP Trunking	Inbound/outbound PSTN ↔ SIP bridge
E. Call Center	CTM browser softphone (WebRTC)	Call handling, disposition, basic CRM view
E. Agent Location	Puerto Vallarta, Jalisco, Mexico	Consumer ISPs (Telmex/Izzi/TotalPlay)
F. Transcription	Groq Whisper Large v3 Turbo	Batch post-call, high accuracy, $0.04/hr
F. Transcription (native)	CTM built-in ASR	Post-call, moderate accuracy
G. CRM	Zoho CRM	Contact records, call history, lead pipeline
H. Case Management	Flow Legal Management	Legal case tracking, intake, call-case linking
I. Data Warehouse	Snowflake	Management reporting from CTM, Zoho, and financial/payments data
J. Payments	TBD	Client payments, billing, collections

14. Vendor Dependencies & Decommissioning Risks

CTM is the hub that all other systems depend on. Removing it requires replacing every integration it provides.



Decommissioning prerequisites (CTM cannot be removed until all are met):

1. ✅ Alternative call routing operational (agent calls connect without CTM)
2. ✅ Call recording with accessible storage (not vendor-locked)
3. ✅ Transcription pipeline producing searchable transcripts
4. ✅ DNI / click tracking / source attribution operational
5. ✅ Zoho CRM receiving call data via new integration
6. ✅ Flow Legal Management receiving call/case data via new integration
7. ✅ Historical recordings migrated from CTM (or acceptance of data loss)
8. ✅ Twilio numbers ported to new carrier (Telnyx) or new numbers provisioned
9. ✅ Snowflake ETL pipelines redirected from CTM/Zoho to new data sources

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15. Cross-References

Document	Purpose
ARCHITECTURE_OPPORTUNITIES.md	Inefficiency analysis and five redesign opportunities
ARCHITECTURE_VISION.md *(forthcoming)*	Target-state RustPBX platform design, technology decisions, and implementation stages
TESTING_PLAN_OF_ACTION.md	Nine-stage testing strategy for the replacement platform
PROJECT_OVERVIEW.md	High-level project summary and RustPBX capabilities